MIL-F-51163A(EA)
17 December 1981
SUPERSEDING
MIL-F-51163(MU)
30 August 1963

MILITARY SPECIFICATION

FILTER UNIT, GAS-PARTICULATE, GED OR EMD, 300 CFM, ABCA-M6A1

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

This specification covers one type of filter unit used to remove chemical agents from air for supply to shelters. The filter unit is either gasoline engine driven (GED) or electric motor driven (EMD) (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry-Cleaning Solvent.
NN-P-530	-	Plywood, Flat Panel.
QQ-S-781	-	Steel Strapping, Flat.
TT-W-572	-	Wood Preservation, Water-Repellent.
PPP-B-636	-	Box, Fiberboard.
PPP-C-843	-	Cushioning Material, Cellulosic.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-B-131 - Barrier Materials, Water Vaporproof, Flexible, Heat-Sealable.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-E-10062 - Engines, Spare or Installed (Other than Aircraft), Preparation for Shipment and Storage of.
MIL-C-17504 - Coating Compound, Acrylic, Clear.
MIL-T-22085 - Tapes, Pressure-Sensitive Adhesive, Preservation and Sealing.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

2.1.2 Other Government documents. The following documents form a part of this specification to the extent specified herein.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

D136-14-152 - Tester, Air Flow, Q54, Assembly.
D150-2-1 - Marking Diagram for Shipping Containers.
TDPL 5-19-1801 - Filter Unit, Gas-Particulate, GED or EMD, 300 CFM, ABCA-M6A1, Assembly.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents that are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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NATIONAL MOTOR FREIGHT CLASSIFICATION COMMITTEE

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited in TDPL 5-19-1801 shall conform to the specifications listed therein or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the filter unit shall conform to the specifications and drawings listed in TDPL 5-19-1801.

3.2 Manufacture and assembly. The filter unit shall be manufactured and assembled in accordance with Drawing D5-19-1801.

3.3 Impeller alignment. The impeller shall not touch the impeller housing in either operational mode when tested as specified in 4.2.4.1.

3.4 Air delivery. The filter unit shall deliver 300 $\pm$ 15 cubic feet per minute at 1.5 inches of water corrected to standard conditions (dry air at 70°F and 29.92 inches of mercury) and 3450 $\pm$ 50 revolutions per minute when tested as specified in 4.2.4.2.

3.5 Workmanship. The filter unit shall be free from damage, such as dents or cracks, and foreign matter, such as oil or grease.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the filter units produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions. However, no more than one lot of gas filters shall be represented in any one lot of finished filter units.

4.2.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure. Sample filter units and the level A packaging shall be examined in accordance with the following classification of defects and MIL-STD-105.

Filter Unit, Gas Particulate, GED (or EMD) 300 CFM, ABCA-M6A1 (D5-19-1801).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Marking incorrect	
103	Damage (see 3.5)	
104	Foreign matter (see 3.5)	

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AQL 0.65 percent defective

105	Impeller alignment	4.2.4.1
106	Air delivery	4.2.4.2

Minor: AQL 2.5 percent defective

201	Protective finish missing or incorrect
202	Marking missing or illegible

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 Impeller alignment. Connect the unit for blow through and operate the unit. Examine the impeller housing for signs that the impeller is touching. Connect the unit for the draw through and repeat the test.

4.2.4.2 Air delivery. Attach inlet and outlet hoses. Measure the volume of air delivered by means of the Q54 Air Flow Tester (D136-14-152).

5. PACKAGING

5.1 Cleaning, drying, preservation and packing.

5.1.1 Level A.

5.1.1.1 Cleaning, drying and preservation. The filter unit, including component parts thereto, shall be cleaned in accordance with Method C-1 of MIL-P-116. Drying shall be accomplished in accordance with one of the drying methods listed in MIL-P-116, provided that the method selected is not injurious to the item. Unless otherwise specified herein, preservations, including the application thereof, shall be accomplished in accordance with the applicable requirements of MIL-P-116.

5.1.1.2 Painted and unpainted metal surfaces. Damaged portions of painted surfaces shall be treated and repainted in accordance with type A of MIL-T-704, using the same color of paint as specified on the drawing. Unpainted exterior noncritical and nonfunctioning surfaces, including threaded area, shall be coated with type P-2 preservative.

5.1.1.3 Metal name and instruction plates, tags and tag waterproofing. These items shall be coated with clear acrylic coating compound conforming to MIL-C-17504. Cloth or paper tags shall conform to the applicable requirements of MIL-STD-129. The above specified coating shall not be applied to the tags until after printing.

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5.1.1.4 Tape application. Painted surfaces to which pressure-sensitive tape is to be applied shall be cleaned with dry cleaning solvent conforming to P-D-680. Unless otherwise specified, pressure-sensitive tape specified herein shall conform to type II of MIL-T-22085. Pressure-sensitive tape applied to painted surfaces shall be coated with acrylic coating compound conforming to MIL-C-17504.

5.1.2 Packaging.

5.1.2.1 GED filter unit and blower assembly. Small openings into the filter unit and blower assembly shall be sealed off with pressure-sensitive tape. Large openings (over two inches in diameter) shall be covered with a metal plate disk or a fitted, 1/4-inch plywood disk, sealing same in place with pressure-sensitive tape. The plywood disk shall be fabricated from material conforming to group A, NN-P-530 (type 1, grade 3 or 4 of CS35). Prior to placing the plywood disk over the opening, the disk surface that will come in contact with metal surfaces shall be faced with class 1 or 2 barrier material, conforming to MIL-B-131. The carrier handles shall be fully retracted and secured in this position with minimum 1-1/2-inch-wide tape, conforming to type IV of PPP-T-97, and this tape shall not be overcoated.

5.1.2.2 Gasoline engine. The air-cooled gasoline engine shall be preserved and packaged in accordance with the type II, level A requirements of MIL-E-10062. The air cooling fins and the head of the air-cooled engine shall be coated with type P-10, grade 2 engine preservative oil.

5.1.2.3 EMD filter unit and blower assembly. The filter unit and blower assembly shall be processed as specified in 5.1.2.1.

5.1.2.4 Electric motor. The electric motor shall be lubricated in accordance with the manufacturer's instructions. The shaft outside the motor and the rigid coupling shall be coated with type P-2 preservative compound. All openings into the electric motor shall be sealed off with pressure-sensitive tape.

5.1.2.5 Air inlet cover and coupling assembly. The air inlet cover and coupling assembly shall be packaged in accordance with Method III of MIL-P-116. Each item shall be individually wrapped with barrier material conforming to type III, grade A, class 2 of MIL-B-121 and taped with tape as specified in 5.1.1.4. The cover shall be nested in the coupling assembly within a style RSC, type CF, V3c fiberboard box conforming to PPP-B-636. Cushioning material conforming to type II, class A or B of PPP-C-843, shall be used to prevent movement of the items within the container. The box shall be closed by taping all seams, corners and the manufacturer's joint, in accordance with the requirements in the appendix of PPP-B-636 for waterproofing weather-resistant style RSC boxes.

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5.1.2.6 Air outlet screen and connector assembly. These items shall be packaged together in accordance with the requirements specified in 5.1.2.5.

5.1.2.7 Elbow coupling. The elbow coupling shall be packaged as specified in 5.1.2.5.

5.1.2.8 Exhaust tube with fittings. The exhaust tube shall be coiled in such a manner as to prevent permanent deformation of the tube. The coil shall then be secured in three or more places with a suitable twine or string. Each fitting shall be wrapped with the same cushioning material as specified in 5.1.2.5 and secured in place with the same tape as specified in 5.1.1.4. The item shall then be packaged in accordance with the requirements specified in 5.1.2.5.

5.1.2.9 Hose, air inlet and outlet with hose clamps. Each length of hose shall be collapsed and placed on the carrier provided for this purpose. The hose and hose clamps shall then be individually packaged as specified in 5.1.2.5.

5.1.2.10 Technical publications. The technical publications shall be packaged in accordance with method 1C-1.

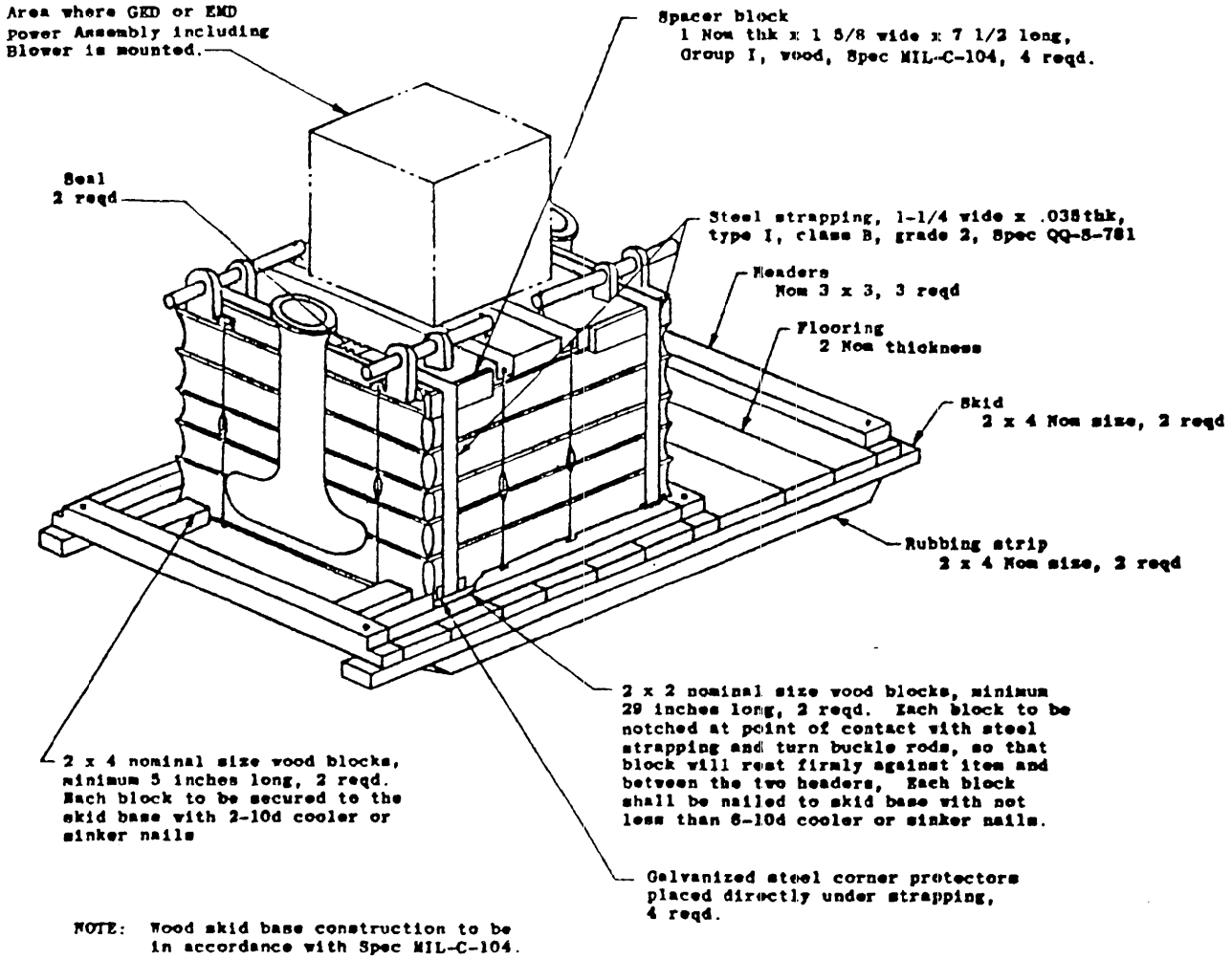
5.1.3 Level C. The GED filter unit and the EMD filter unit and their components, repair parts, and technical publications shall be individually preserved and packaged to provide adequate protection against corrosion, deterioration and physical damage from the supply source to the first receiving activity for immediate use.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 GED filter unit and EMD filter unit. These items, including components, repair parts, accessories, tools and technical publications, preserved and packaged as specified in 5.1.2.1 and 5.1.2.3 through 5.1.2.10, shall be packed in a type II, class 1 or 2, style A, wood crate, conforming to MIL-C-104. The filter unit shall be placed on the wood skid base in the manner shown in figure 1, blocking and tying down the item as indicated thereon. The engine and blower assembly that is mounted to the top of the filter unit shall be located in such a position that the blower air outlet opening will be in close proximity to the air inlet opening on the filter unit and not protrude over the side of the filter unit. Unit packages, containing components, repair parts, accessories, and technical publications, shall be secured to the skid base of the crate with galvanized strapping, conforming to QQ-S-781; or these items may be secured to the equipment with tape in a location that will not cause damage to any of the mounted components on the filter unit. Tape used to secure these packaged items to the equipment shall be minimum 1-1/2 inches wide, conforming to type IV of PPP-T-97, and shall not be overcoated.

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NOTE: Wood skid base construction to be in accordance with Spec MIL-C-104.

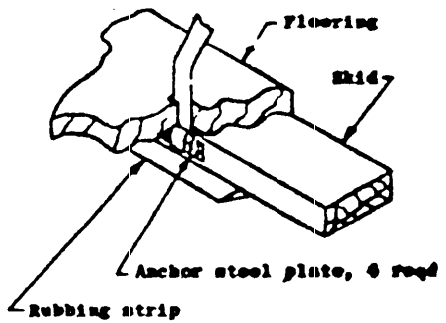


Figure 1. Tie-Down Arrangement, Filter Unit

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5.2.2 Level C.

5.2.2.1 GED and EMD filter units. Each filter unit and its respective components, repair parts, and technical publications, preserved and packaged as specified in 5.1.3, shall be packed to insure carrier acceptance and safe delivery at the first destination for immediate use. Containers shall be in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules. The crate or the finished wood parts thereof shall be completely immersed for a minimum of three minutes in a water-repellent, wood preservative, conforming to type II, composition A of TT-W-572.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with D150-2-1.

6. NOTES

6.1 Intended use. The filter unit covered by this specification is intended for use in providing filtered air to pressurized protective shelters.

6.2 Ordering data. Acquisition documents should specify:

(a) Title, number and date of this specification.
(b) Gasoline engine driven or electric motor driven.
(c) Level of packaging and packing.

6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement, QSTAG - 296. When amendment, revision, or cancellation of the specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.4 Component specifications. Component specifications for the ABCA-M6A1 filter unit are:

Title	Number
Filter, Particulate, 150 cfm, M9A1	MIL-F-51165
Filter, Gas, 150 cfm, M10	MIL-F-11137

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A657

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-F-51163A(EA) - FILTER UNIT, GAS-PARTICULATE, GED or EMD, 300 CFM, ABCA-M6A1

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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